

Writing a scientific report

In this document we outline some of the points that we noticed during the first programming assignments. In science, it is important to be able to communicate your findings to your peers in limited space (both in written form as well as presentation form). Such reporting should often be precise. Note that during a scientific master, you are expected to do experimentations and discuss the findings in a scientific way. Hereby we give some pointers on how to go about assignments that include such a scientific report. These guidelines are aimed to aid you in the reporting phase but do not need to be explicitly followed.

Some additional pointers:

- <https://bettersci.wordpress.com/> (blog by Holger Hoos)
- <https://www.academic-skills.org/> (website maintained by Thomas Moerland)

Code:

- Note that you will be graded mostly on the report.
 - That means that the report should be of high quality, and convey important information about the concepts that you have implemented.
 - That does not mean that the code can be of low quality, as the person grading the code should be able to run the code / verify it's working (possibly through the use of unit tests)
- If a class and function framework is provided, make use of this structure (so that easy testing can be facilitated)
- Ensure that the code is produced by yourself, avoid making use of large language models.
- Make use of appropriate libraries when possible; of course, refrain from using libraries that contain the solution already.
- Ensure that code is well documented, both at function level and at global level.

Report (structure):

- Introduction
 - This often includes motivation, problem statements and research questions
- Background/Theory
 - minimal, should be functional to support
 - Please cite relevant work.
 - May be in the introduction
- Experiments/Methods
 - Explain the implementation of the algorithms and experiments.
 - What is your hypothesis?
 - What metrics are chosen to evaluate the experiments?

- Discussion
 - What did your experiments reveal in view of your research question, how did this compare to your expectations
- Conclusion
 - There should be no new insights, just recap and answer the research question

Report (content):

- Handling limited space:
 - Ensure to have the text to the point (concise but covering the necessary points)
 - Latex supports multiple small figures/tables next to each other
 - Consider carefully what text is required → Having little space typically requires more reflection
 - Space is typically restrictive therefore provide a high-level explanation of methods/algorithms used (be concise).
 - Think carefully on what type of results should be included → consider what helps you present your conclusions and infer/analyse accordingly
- Plots:
 - The plots should be designed in a way that it is easy to read and understand.
 - Include axis titles, title of plot and each figure should be accompanied by a clear caption that makes the plot self-contained.
 - Each figure should be referenced in the report.
 - Axis should align if multiple plots are to be compared. If there is space, put multiple curves into one plot, makes comparison easier (but not too many)
 - Add confidence bounds when averaging across multiple seeds.
 - Metrics should be stated explicitly.
- Language
 - When writing, try not to write everything that comes to mind. Try to write concisely, not adding statements that do not bring additional value. Text often requires several revisions before it reaches its final state.
 - Make sure to use scientific / formal language. Active sentences are often preferred in the field of Computer Science, as they read easier than passive sentences (but not required).
 - Do not refer back to Python function names, this can be done in the README.md
- Please state values for chosen hyperparameter values, these could have a large impact on the results (e.g. starting configs in SMBO)
- Also: always state key assumptions, including how the question / assignment is interpreted if you think there is ambiguity (or mistakes)
- Do not make hard conclusions about matters, without presenting evidence to support this